ONNX: Helping Developers Choose the Right Framework

ONNX or Open Neural Network Exchange (onnx.ai) is a community project created by Facebook and Microsoft. It is intended to provide interoperability within the AI tools community. ONNX unlocks the framework dependency for AI models by bringing in a new common representation for any model, which allows easy conversion of a model from one framework to another.

Deep learning with neural networks is accomplished through computation over dataflow graphs. Developers use frameworks such as CNTK, Caffe2, Theano, TensorFlow, PyTorch and Chainer to represent a computational graph. All these provide interfaces that make it simple for developers to construct computation graphs and runtimes that process the graphs in an optimised way. The graph serves as an intermediate representation (IR) that captures the specific intent of the developer's source code, and is conducive for optimisation and translation to run on specific devices (CPU, GPU, FPGA, etc).

The pain point with many frameworks is that each one of them follows its own representation of a graph with similar capabilities. In simple words, a model is bound to a framework with the stack of API, graph and runtime. We can't take a model from one framework to another. Besides this, frameworks are typically optimised for certain specific characteristics, such as fast training, supporting complicated network architectures, inference on mobile devices, etc. It's up to the developer to select a framework that is optimised for one of these characteristics. Additionally, these optimisations may be better suited for particular stages of development. This leads to significant delays between research and production due to the necessity of conversion.

ONNX was launched with the goal of democratising AI, by empowering developers to select the framework that works best for their project, at any stage of development or deployment. The Open Neural Network Exchange (ONNX) format is a common IR to help establish this powerful ecosystem. By providing a common representation of the computation graph, ONNX helps developers choose the right framework for their task, allows authors to focus on innovative enhancements, and enables hardware vendors to streamline optimisations for their platforms.

ONNX provides the definition of an extensible computation graph model, as well as definitions for built-in operators. Each computation graph is structured as a list of nodes forming an acyclic graph. Here, the node is defined as a piece of a computational block with one or more inputs and one or more outputs. Each node is a call to an operation. The graph also has metadata to document or describe the model. Operators are implemented externally to the graph, but the built-in operators are portable across frameworks. Every framework supporting ONNX will provide implementations of these operators on the applicable data types.

As of today, ONNX supports various frameworks like Caffe2, Chainer, Cognitive toolkit, MxNet, PyTorch, PaddlePaddle, MATLAB and SAS. ONNX also supports